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An Efficient Sentiment Analysis Approach for Product Review using Turney Algorithm

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Abstract

Sentiment analysis can be done by means of Classification and its most important tasks are text categorization, tone recognition, image classification etc. Mostly the extant methods of supervised classification are based on traditional statistics, which can provide ideal results. The main aim is to increase the accuracy and to report the manufacturer about the negatives of the product. The major problem is categorization of sentiment polarity, which is the problem of sentiment analysis. There are two levels of categorization and they are Review-level Categorization and Sentence-level Categorization. Categorization of review-level becomes arduous when we attempt to classify the reviews respect with their specific rating related to star-scaled. Second, Review-level Categorization has a drawback in Implicit-level sentiment analysis. Mostly SVM, Naïve Bayesian and Decision Tree are mainly used to improve the efficiency of classification. Amazon Dataset is used as Dataset in proposed system to improve the accuracy of Turney algorithm. Semantic Orientation (SO) with Point wise Mutual Information yields good results than other classification methods. The review level gets subjected as positive value, on acquaintance of positive average SO. On the other hand, the review level acquires a negative level in accordance with attainment of negative average SO.

Keywords: Pointwise Mutual Information;Semantic Orientation; Sentimental Analysis

1. Introduction

Now-a-days Sentiment analysis is used frequently to analyze the customer feedback. Customer opinion is more important for the success of the product. In olden days people hear reviews about the product and then, they decide

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the quality of the product [1]. This sentiment analysis is extensively used in social media. Natural Language Processing (NLP) gets employed to work out the concept of sentiment analysis also entitled as opinion mining or emotion AI as it's far and wide notion that eventually gets identified as say-so aspect of customers'. Datasets utilized are manipulated on the basis of textual analysis procedure as well as Computational linguistics in order to excavate intrinsic information available [2].

There are two types of sentences: Subjective and Objective sentences. Mostly subjective sentence contains more sentiment than objective sentences. For example, Objective sentence: Milk is white and Subjective sentence: Milk tastes good. The subjectivity of words depends on the context of the words. Objective document may also contain subjective sentence. To overcome these difficulties, we implement proposed system with Turney algorithm which also increases the accuracy of the reviews. Here, we use Recall, F-measure and Precision parameters as metrics to assess the accuracy. Precision gives good accuracy than others. Finally, the system calculates the Polarity of the reviews. If the reviews contain more negatives, our system will recommend the manufacturer to recover the negatives in their products during their forthcoming manufacturing process to increase their sales [6,7]. This paper presents a classification algorithm in order to classify the reviews as either negative or positive. This method takes the input of Amazon dataset and gives the output as positive or negative review which is given by the user. First, the dataset is pre-processed then feature is extracted using POS Tagging with a count value. For the most portion of technically necessitated information can be chiefly acquired from adjective and adverb segment. These two compositions in parts of speech comparatively serves better than other segments available in a sentence. Point wise Mutual Information (PMI) parameter accompanied with SO is assessed by means of implying the phrase extorted [10]. This evaluation merely imparts the correlation existing among words being processed so far. Latterly, those processed linguistic terms are semantically categorized and classified by means of the review attained. A Positive review gets evolved on attainment of a positive average semantic orientation. Otherwise, it gets hold for a negative semantic orientation.

2. Related Work

Durgesh K implemented three different types of classification techniques such as Naïve Bayesian, Random Forest and Support Vector Machine (SVM). Random Forest yields better results than other two methods. The author uses Twitter dataset to classify the review. He selected four subsets from a complete set namely A, B, C and D that possibly comprises of 300, 3,000, 30,000 and 300,000 vector count correspondingly. F1 scores are obtained with these sets of vector. For Review-level categorization F1 scores obtained is fairly low over 0.73. There is also a problem of neutral sentence, where judgment is difficult [3].

Peter D implemented Turney the unsupervised classified algorithm for thumbs up and thumbs down semantic orientation. He conducted experiments with 410 reviews of E-opinions which gives an average accuracy of 74%. For movie reviews it is difficult to predict and it produces an average accuracy of about 66%. 80% to 84% of average accuracy is achieved for travel reviews. This paper talks of E-opinion reviews of classification using thumbs up and thumbs down algorithm [4].

3. Classification Methods

SVM, Naïve Bayesian, Decision tree are well known classification methods. SVM is used for text categorization. Hyper plane is represented using Vector. Hyper plane separates the vectors from one class to another class. By this way classification is done in SVM. In Naïve Bayesian we use Bayes rule to classify the text. It is also used for text categorization. Decision tree can be easily done using J48 classifier in Weka package. All these give good results of accuracy [5].

3.1 SVM

The practice of classification can be proficiently carried out by means employing Support Vector Machine (SVM) methodology on the information available. Eventually, all these data available on the whole is set apart as linear and non-linear segregations. Furthermore, linear kernel (an optimal hyper plane that segregates linearly) is been searched for on achieving a robust classification accuracy after employing SVM methodology [6, 21]. This linear kernel optimally serves as a decision boundary for isolating a specific set of information available in a class from those available devoid of any correlation. Mathematically the aforementioned hyper plane gets designated as, W . X+b = 0. Here, the vector weight gets specified as W that perhaps ranges from, w_1, w_2,...,w_n. Training of information can be accomplished by means of manipulating tuple X accompanied by a scalar component 'b'. Furthermore, minimized value of ‖W‖ gets evaluated by means of computing

$$\sum_{i=1}^{n} \alpha_i y_i x_i \tag{1}$$

where support vectors (X_i) labels are particularized on yi and numerical factors gets specified as α_i.

$$y_i=1 \text{ then } \sum_{i=1}^{n} w_i x_i \geq 1 \tag{2}$$

$$\text{if } y_i=-1 \text{ then } \sum_{i=1}^{n} w_i x_i \geq 1 \tag{3}$$

On unproductive segregation by means of employing linear manner, SVM transmutes the procedure of classification by utilizing non-linear way of mapping. Certainly this procedural alteration completely reveals a higher dimension form from the data being processed. For such a profound form of procedural transformation a kernel function called Gaussian Radial Basis Function (RBF) is absolutely utilized in order to extort most robust form of results [7,8].

$$K(X_i, X_j) = e^{-y[X_i - X_j]^2} / 2 \tag{4}$$

Where X_i are support vectors, γ is a free parameter and X_j are testing tuples.

3.2 Naive Bayes

D confines to be the dataset that comprises features that scales up to n-dimensional form of feature vector specified as $X=x_1,x_2,..,x_n$, that are to be trained on the basis of characterized tuple set. Here, "n" accounts for dimensions specified on every particular tuple. Initially the classifier imposed concentrates on 'm' number of classes ranging from $C_1,C_2,...,C_m$ [9]. The tuple X of data considered is liable to get categorized into any class C_i after classification when, $P(C_i|X) > P(C_j|X)$, where $i,j \in [1,m]$and $i \neq j$. Furthermore,

$$P(C_i \mid X) = \prod_{i=1}^{n} P(x_i \mid C_i) \tag{5}$$

3.3 Decision tree

All classified form of information can be capably signified by means of utilizing decision tree that suitably keeps apart those components comprises of features acquired out of finite distinct domains. Classified features of every distinct domain are personified into discrete classes. Every single feature observed, probably that different from one another serves to be the internal nodes of this decision tree [10, 11, 12]. All those other data coming under the similar domain also gets appended to the former one. Finally, all sorts of data get labeled under a specific class that possibly got classified as leaf node or any of the internal nodes belonging to the decision tree. Moreover, amalgamation of arithmetical as well as computing methodologies compactly yielded a robust methodical procedure named as decision tree. Henceforth, this structure can resourcefully systematize those data being processed by it in a categorical manner [13, 14, 16].The information to be categorized progresses as,

$$(x, Y)=(x_1,x_2,x_3,\ldots\ldots,x_k,Y) \tag{6}$$

The elements available in (6) gets designated as, reliant variable that is also named as target variable Y. x gets specified as input variable that typically encompasses x_1, x_2, x_3….xn [15, 17, 18].

4. Proposed Algorithm

Our proposed algorithm is based on turney algorithm. The flow chart shown in Fig.1. depicts about proposed process of our algorithm. Firstly, review data that is Amazon Dataset is given as input dataset. PoS Tagging is done to extract feature and then semantic orientation is performed. Nature of review utterly relies upon the outcome of average value of SO parameter. Polarity gets indicated as positive only when average becomes positive.

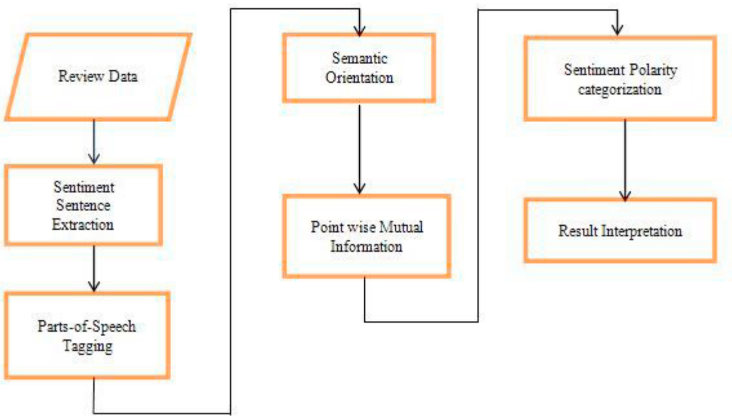

Fig.1. Flow diagram for proposed system

4.1 Selection of Phrase and feature phrase extraction

The first step in classification is pre-processing which removes unwanted data and stop words. The stop word removes based on stop word list file. Stop words do not affect the meaning of the sentence. Some of the stop words "a", "the", "of" which reduces the corpus size without losing any information. There are 8 parts of speech form which two words are extracted (adjective and adverb). Some words can have two different meaning at two different places. For example, "great book" for book review is positive and "great flop" for movie review is negative. Table 1 shows the Pos tag and their meaning.

Table 1. Pos Tags and their meaning

PoS Tagging	Meaning
JJ	Adjective
RB, RBR, or RBS	Adverb
NN, NNS	Noun
VB, VBD, VBN or VBG	Verb
CC	Cardinal Number

Tag patterns are extracted from two word phrase from the review. Semantic Orientation is done with these kinds of patterns.

Table 2. Two-Word Phrase

First Word	Second Word	Third Word
JJ	NN or NNS	Anything
RB, RBR, or RBS	JJ	Not NN or NNS
JJ	JJ	Not NN or NNS
RB, RBR, or RBS	VB, VBD, or VBN	Anything
NN, NNS	JJ	Not NN or NNS

4.2 Seed words

We should find the seed words and we want to pass it to the algorithm to find the semantic orientation. If the phrase occurs along with negative seed word, the phrase is negative. If the phrase occurs along with positive seed word, the phrase is positive.

Table 3. Seed Words

Positive Seed Words	Excellent, Best, Good
Negative Seed Words	Poor, Bad, Hate, Terrible

4.3 PMI and SO

The Point wise correlative information between two words (extracted word and seed word) can be given as

$$PMI(word_1, word_2) = \log_2\left(\frac{p(word_1 \,\&\, word_2)}{p(word_1)\,p(word_2)}\right) \tag{7}$$

Here, p(word1,word2) gives the probability of the word co-occur. Here, p(word1) and p(word2) gives the probability of the word occurs independently. The ratio between p(word1,word2) and p(word1) p(word2) gives the degree of statistical dependence between those words. After the calculation of PMI we have to calculate the semantic orientation. Thus, SO is calculated as

$$\text{SO(phrase)} = \text{PMI(phrase, \{positive paradigms\})} - \text{PMI(phrase,\{negative paradigms\})} \tag{8}$$

If the phrase is with positive seed word, then that phrase is considered as positive. If the phrase is with negative seed word, then the phrase is considered as negative. PMI-IR issues query to search engine to find the number of hits (matching document). The AltaVista NEAR operator is used to search documents which is more efficient than AND operator. From equation (2) and (3) we can derive this equation with NEAR operator [10].

$$\text{SO(phrase)}=\log_2\left(\frac{hits(phraseNEAR\text{"}excellent\text{"})hits(\text{"}poor\text{"})}{hits(phraseNEAR\text{"}poor\text{"})hits(\text{"}excellent\text{"})}\right) \tag{9}$$

p_query = (good OR best OR….OR superior)
n_query = (bad OR poor OR…..OR inferior)
This can give positive semantic orientation or negative semantic orientation.

4.4 Sentiment Classification

After finding semantic orientation we have to find the average semantic orientation. If the central semantic orientation is negative, the review is negative and the review is recommended review. In case there is an attainment of a negatively polarized average SO then the review does not serves to be an optional review and will not be encouraged to be induced anymore [19,20].

5. Experiments

Experiments are done with 5000 labeled positive and negative product reviews. It consists of reviews of the products. There are 2500 labeled positive reviews and there are 2500 labeled negative reviews. Turney algorithm gives better results.

Table 4. Performance Analysis of differ Classifier Algorithms

Algorithm	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC
Turney	0.781	0.213	0.788	0.787	0.787	0.871
Decision Tree	0.607	0.393	0.607	0.607	0.607	0.611
SVM	0.696	0.304	0.731	0.696	0.684	0.696
Naive Bayesian	0.781	0.212	0.781	0.782	0.780	0.870

6. Results

The Table 4. shows the performance analysis of different classifiers which is measured using precision, recall, f-measure and roc where turney classifier yields the good accuracy (0.871) than other classifiers like Decision tree, Naïve Bayes and SVM for movie review dataset and its graphical representation is been shown in Fig.2.. Decision tree yields good accuracy with roc (0.611). SVM yields good accuracy with precision (0.731). Naïve Bayes yields good accuracy with roc (0.870).

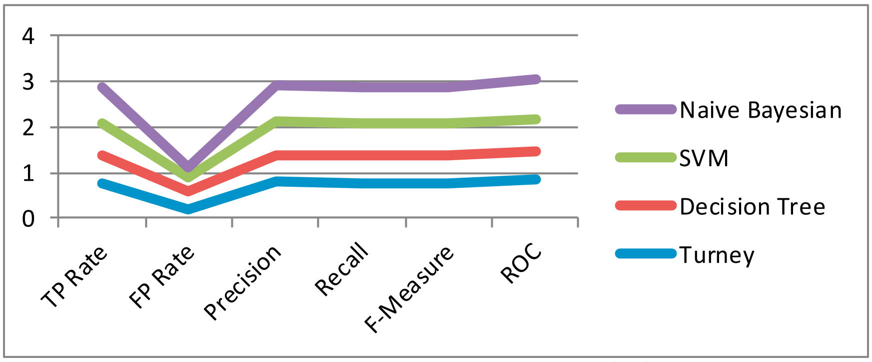

Fig.2. Performance Analysis different classifier Conclusion

Thus, the classifiers Decision tree, SVM, Naïve Bayes yields less accuracy we go for Semantic Orientation is used to classify the reviews. Semantic Orientation with PMI increases the performance of the classifier. SO-PMI is simple, unsupervised, simple to implement and it doesn't restrict to adjectives, but it requires a large corpus. The PMI algorithm has three simple steps: first is to extracts two-word phrase containing adjective and adverbs. Second is to find Semantic Orientation of phrases and third is to take average of all SO and assign sentiment as positive or negative. Turney classifier algorithm yields good performance measure than the other three algorithms for the Benchmark Dataset. Turney classifier classifies accurately upto ~78% of accuracy where the other three algorithms classifies upto ~69%.

7. Future Work

The errors and the negatives of the product can be mailed automatically to the manufacturer to rectify their errors. They can also increase their sales in the market. In future work, to improve the accuracy we can use semantic orientation with other features in a combined manner as like a combination in a supervised classification algorithm.

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